Polarization of the Sky and Estimation of the Sun's Position

Asservissement visuelle - 3A ROB

Fauzi AKBAR

ENSTA Bretagne

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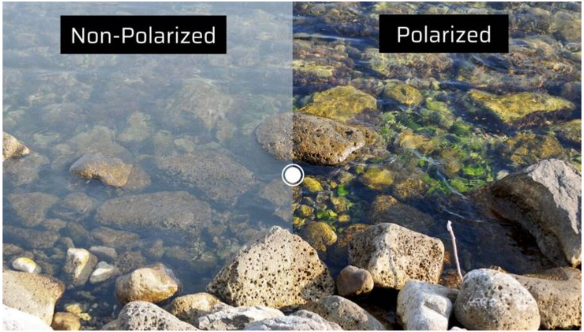
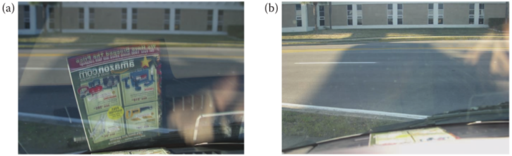
What is Polarization?

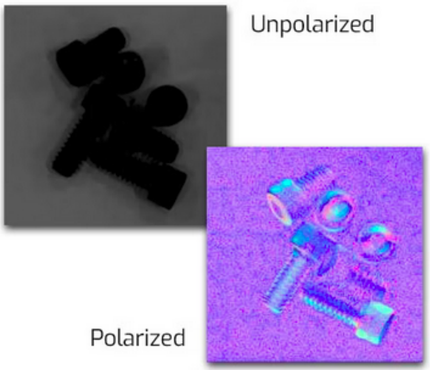


Example: Privacy Screen Protector and Polarized Sunglasses.

Taking Advantage of Polarization

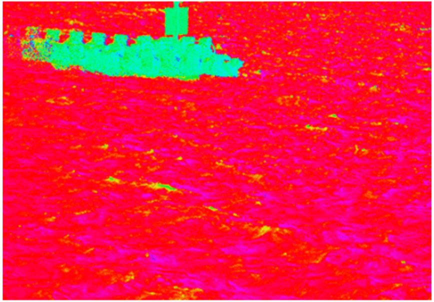


View from vehicle as seen (a) without polarized sunglasses and (b) with polarized sunglasses. (Photo courtesy of D. H. Goldstein.)



Reduce the reflection | Enhance the contrast

Object Detection



Raw photo (left) and Polarized Camera (Right)

Navigation


AAAS
Science
nature electronics
REUTERS
Le Monde
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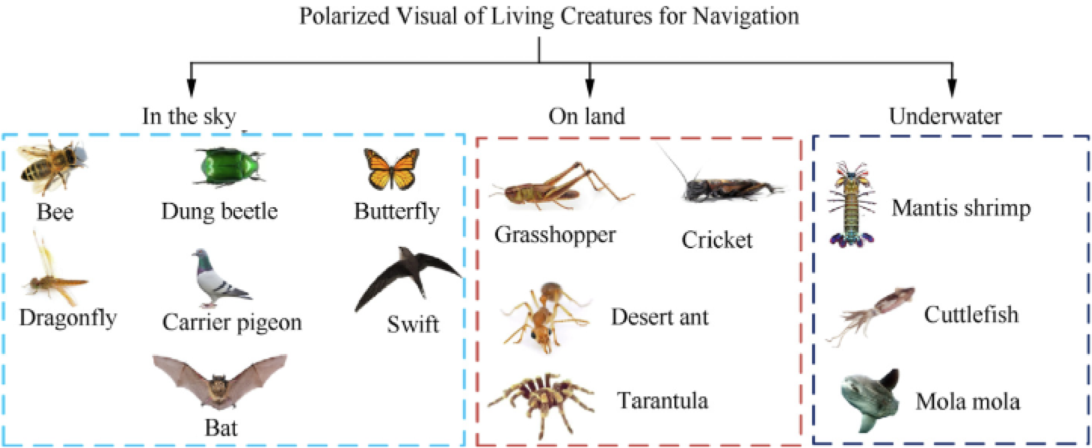
Bio-Inspired Navigation



The Celestial Compass: Bio-Inspired Polarized Light Navigation

Insects navigate using the polarization pattern of the sky.

Key Biological Insight

Many insects extract the **angle of polarization (AoP)** of skylight using a specialized region of the eye called the **Dorsal Rim Area (DRA)**. This forms a natural **celestial compass**.

- DRA detects e-vector orientation from Rayleigh-scattered skylight.
- Neural circuits compare AoP to an internal sky-polarization map.
- Insects infer Sun azimuth even when the Sun is not visible.
- Basis for robust bio-inspired navigation sensors.

How the Celestial Compass Works

From the Dorsal Rim Area (DRA) → Neural Processing → Navigation

1. Dorsal Rim Area (DRA)

- Specialized ommatidia at top of the compound eye.
- Microvilli arranged in orthogonal directions.
- Extremely sensitive to polarization (AoP, DoP).

2. POL Neurons

- Receive DRA signals (lamina → medulla).
- Each neuron tuned to a preferred e-vector angle.
- Construct a neural **polarization pattern map**.

How the Celestial Compass Works

From the Dorsal Rim Area (DRA) → Neural Processing → Navigation

3. Central Complex (Compass Center)

- Combines AoP map with time-of-day.
- Computes Sun azimuth using an internal sky model.
- Outputs a stable **compass heading**.

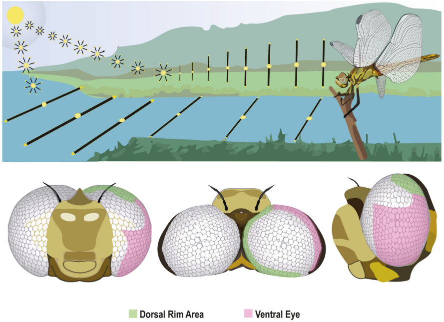
4. Navigation Behavior

- Path integration (desert ants).
- Long-distance foraging (bees).
- Straight-line escape behavior (dung beetles).

This mechanism is widely used for bio-inspired robot navigation.

Dorsal Rim Area (DRA) and Its Artificial Equivalent

DRA as a Biological Polarization Sensor



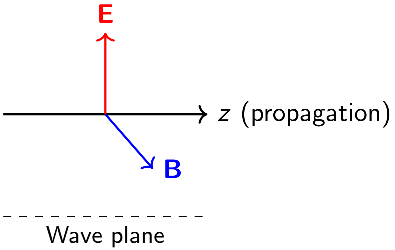
Biological System	Artificial Equivalent	Function
Dorsal Rim Area (DRA)	Polarization camera / polarimeter	Measures AoP and DoP of skylight
POL neurons	Signal processing / filter banks	Extract e-vector orientation
Central Complex (compass center)	Navigation model / sun compass algorithm	Computes absolute heading
Insect movement (path integration)	Robot heading and control system	Executes navigation

Going back to Polarization, What is light?

- Light = electromagnetic wave (EM)
- Consisting of electric field **E** and magnetic field **B**
- Propagates in a vacuum at the speed of:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3 \times 10^8 \text{ m/s}$$

- **Transverse**: **E** $\perp$ **B** $\perp$ propagation direction



Maxwell's Equations in Vacuum Space

Without load & current ($\rho = 0$, $\mathbf{J} = 0$)

$$\nabla \cdot \mathbf{E} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \text{(Faraday)}$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad \text{(Ampère-Maxwell)}$$

Question: How these equations generate **waves**?

Derivation: The Alembert equation for $\mathbf{E}$

1. Take `curl` from Faraday:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B})$$

2. Vector identity + $\nabla \cdot \mathbf{E} = 0$:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$$

3. Substitution with Ampère-Maxwell:

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad \Rightarrow \quad -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

4. Finally:

$$\boxed{\nabla^2 \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0}$$

$$\boxed{\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0} \quad \text{with} \quad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

Wave Equation For **B**

Similar steps:

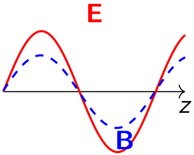
- Take `curl` from Ampère-Maxwell
- Use The Faraday Law
- Get:

$$\boxed{\nabla^2 \mathbf{B} - \frac{1}{c^2}\frac{\partial^2 \mathbf{B}}{\partial t^2} = 0}$$

Plane-Wave Solution

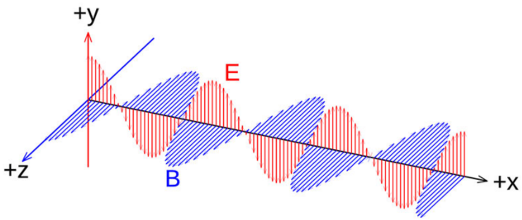
$$\mathbf{E}(z,t) = \mathbf{E}_0 \cos(kz - \omega t) \quad ; \quad \mathbf{E}_0 \perp \hat{z}$$

$$\mathbf{B}(z,t) = \frac{1}{c}\hat{z} \times \mathbf{E}(z,t)$$



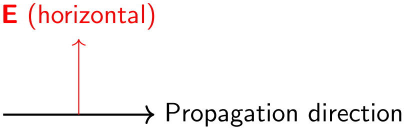
- $\omega = ck \Rightarrow$ dispersion relationship
- $\mathbf{E}_0$ determine polarization

Plane-Wave Solution



How?

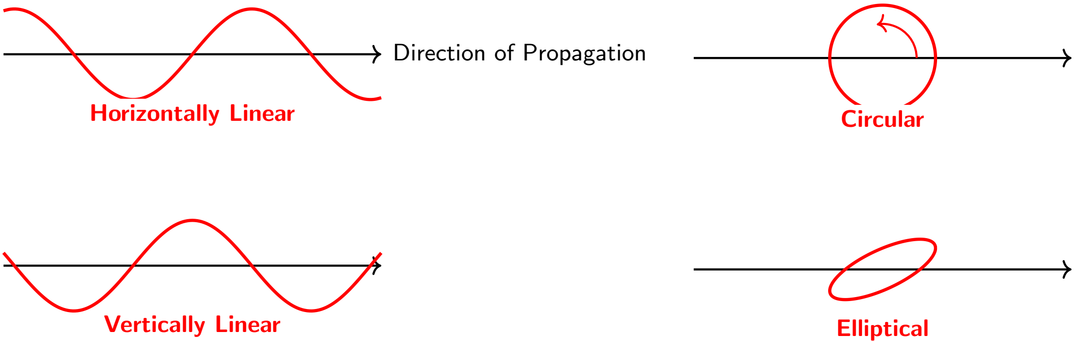
- **E** is vector → the direction of oscillation can remain constant
- If direction of **E** is random → unpolarized
- If direction of **E** is constant → linearly polarized



Horizontal linear polarization

Types of Polarization: Visualization

Direction of Electric Field Oscillation E



The wave **E** *is always perpendicular to the direction of propagation.*

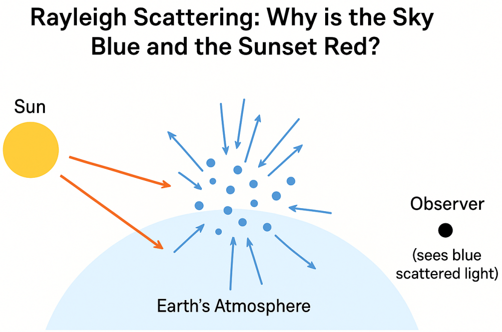
Types of Polarization: Classification

The Four Main Types of Polarization

- **Linear**: **E** oscillates in a single fixed direction
  $\rightarrow$ Used in Polaroid sunglasses, LCD screens
- **Circular**: **E** rotates (right-handed or left-handed)
  $\rightarrow$ Used in 3D glasses
- **Elliptical**: Combination of linear components with a phase difference
  $\rightarrow$ Found in optical fibers
- **Unpolarized**: **E** has random directions
  $\rightarrow$ Produced by direct sunlight

In all cases: $\mathbf{E} \perp \mathbf{B} \perp$ direction of propagation

Rayleigh Scattering: Visualization



Shorter wavelengths (blue) scatter more strongly than longer ones (red).

Rayleigh Scattering: Concept and Formula

Scattering of Light by Small Particles

Basic Idea

When light interacts with particles **much smaller than its wavelength** (like air molecules), it is scattered in different directions.

Consequences

- Shorter wavelengths (blue) scatter much more strongly than longer ones (red)
- The sky appears blue during the day
- Sunsets and sunrises look red because blue light is scattered out

Scattering explains the blue sky and red sunsets.

Skylight as Partially Polarized Light

Sunlight becomes partially polarized after Rayleigh scattering.

Key Idea

In Rayleigh scattering, each air molecule scatters light more efficiently in directions **perpendicular** to the incident electric field. This produces **partial linear polarization** of skylight.

- At 90° from the Sun, the scattered light is **most polarized**.
- At 0° or 180° (toward or opposite the Sun), polarization is minimal.
- The light we see from the sky is a **mixture of polarized and unpolarized components**.

Stokes Parameters and the Stokes Vector

A complete description of any state of polarization.

Stokes Parameters

$S_0 = I_H + I_V$ (Total intensity)

$S_1 = I_H - I_V$ (Horizontal vs. vertical linear)

$S_2 = I_{+45} - I_{-45}$ (Diagonal linear)

$S_3 = I_R - I_L$ (Right vs. left circular)

Stokes Vector Representation

$$\mathbf{S} = \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix}$$

- Fully polarized: $S_0^2 = S_1^2 + S_2^2 + S_3^2$
- Partially polarized: $S_0^2 > S_1^2 + S_2^2 + S_3^2$

The Stokes formalism unifies intensity and polarization information.

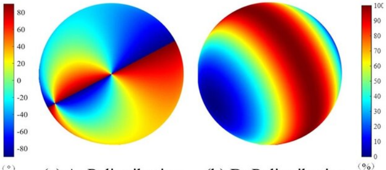
Polarization Pattern of the Sky

Angle and Degree of Polarization in Skylight

Derived Quantities

$$\text{Degree of Polarization (DoP)} = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0}$$

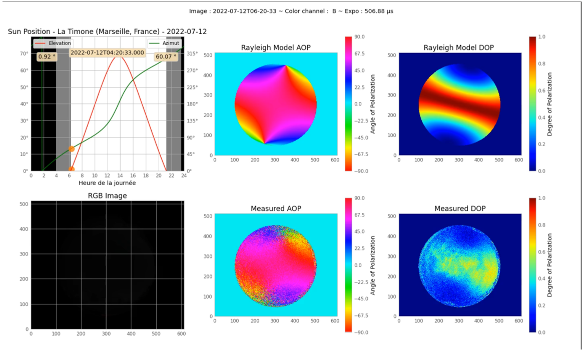
$$\text{Angle of Polarization (AoP)} = \frac{1}{2}\tan^{-1}\left(\frac{S_2}{S_1}\right)$$



(a) AoP distribution (b) DoP distribution

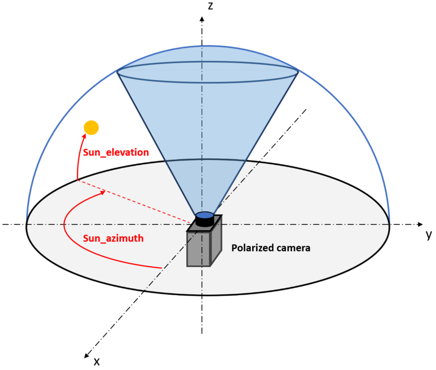
Source: Y. Fan et al., "Adaptation Analysis of Polarized Light Orientation Methods based on Rayleigh Model," IEEE

Polarization Pattern of the Sky (click to play the video)



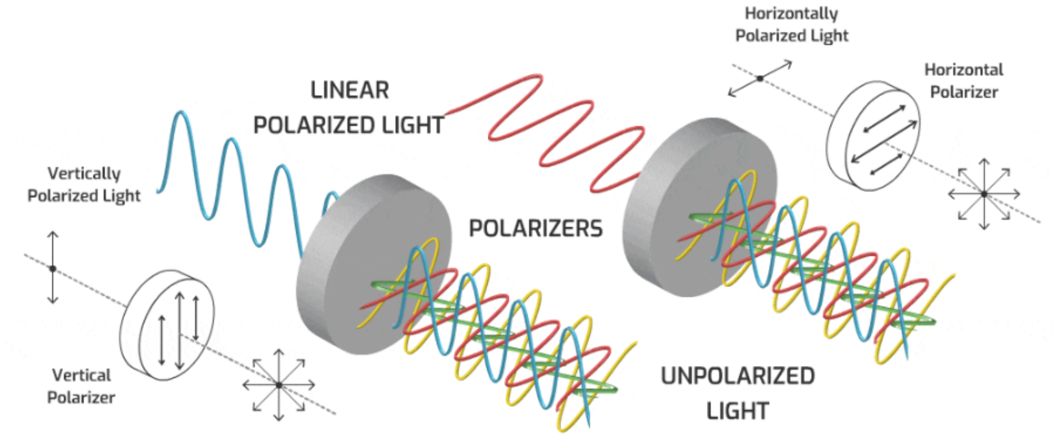
Poughon, Léo, et al. "A stand-alone polarimetric acquisition system for producing a long-term skylight dataset." 2023 IEEE SENSORS. IEEE, 2023.

OpenSky: Sun Polarization Simulator



Moutenet, A., Poughon, L., Toulon, B., Serres, J. R., Viollet, S. (2024). Opensky: a modular and open-source simulator of sky polarization measurements. IEEE Transactions on Instrumentation and Measurement, 73, 1-16.

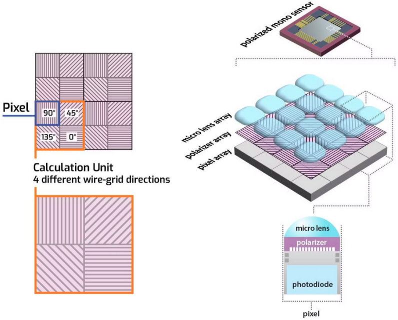
Sunlight Polarization With Polarizer



Sunlight Polarization With Polarizer (click to play the video)

Polarization Camera

An example : PHX050S1-QC-IX LUCID Camera



Compute Stokes from four polarizer angles

Denote measured intensities: $I_0, I_{45}, I_{90}, I_{135}$ (each is the image of pixels under that micro-polarizer).

$$S_0 = I_0 + I_{90} \quad (\text{or } I_{45} + I_{135})$$

$$S_1 = I_0 - I_{90}$$

$$S_2 = I_{45} - I_{135}$$

(This is directly from the linear equations for those four angles.)

CalculatAoP and DoP

Derived quantities

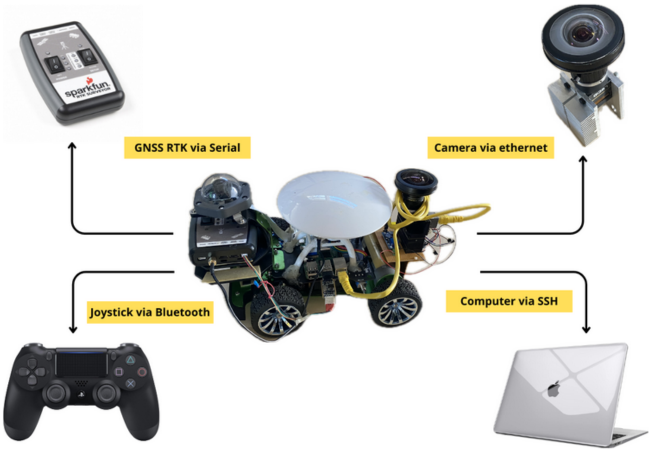
$$\mathrm{DoP} = \frac{\sqrt{S_1^2 + S_2^2}}{S_0}$$

$$\mathrm{AoP} = \tfrac{1}{2}\,\mathrm{atan2}(S_2, S_1)$$

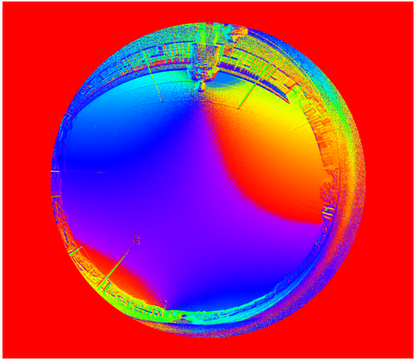
Example Python (numpy) — compute AoP & DoP

```
# raw_mosaic: 2D numpy array from camera sensor for 2x2 repeating pattern:
#  [ 90°  45°
#    135° 0° ]
import numpy as np

def demosaic_4angles(raw):
    h,w = raw.shape
    I90  = raw[0:h:2, 0:w:2].astype(float)   # 90°
    I45  = raw[0:h:2, 1:w:2].astype(float)   # 45°
    I135 = raw[1:h:2, 0:w:2].astype(float)   # 135°
    I0   = raw[1:h:2, 1:w:2].astype(float)   # 0°
    return I0, I45, I90, I135

I0,I45,I90,I135 = demosaic_4angles(raw_mosaic)

S0 = I0 + I90
S1 = I0 - I90
S2 = I45 - I135

eps = 1e-8
DoP = np.sqrt(S1**2 + S2**2) / (S0 + eps)
AoP = 0.5 * np.arctan2(S2, S1)    # radians, range (-pi/2, pi/2)
```

Robot with Polarization Camera

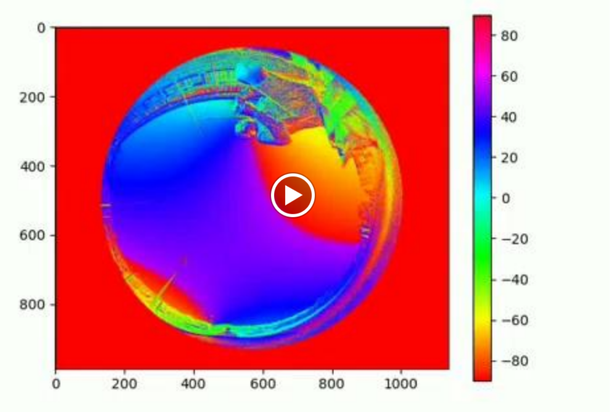


Camera Output

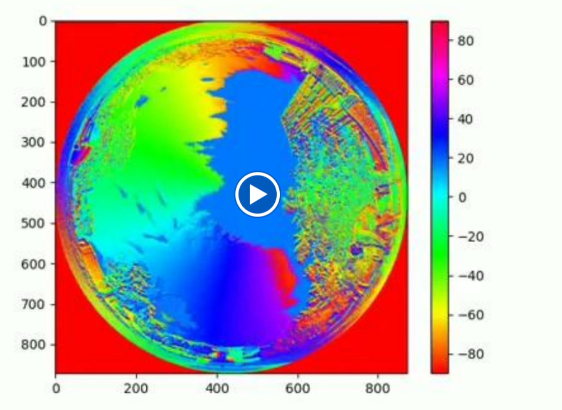


Camera Output for Angle of Polarization

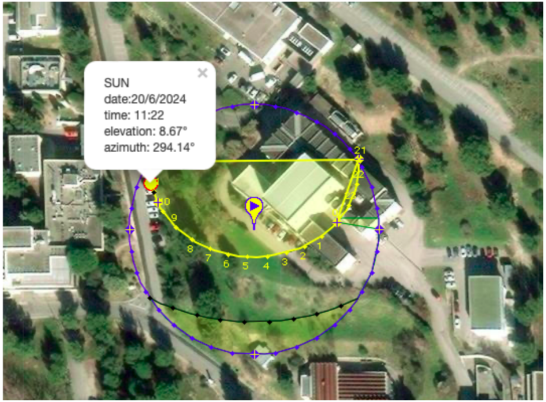
Robot Experiment to get Dataset (click to play the video)



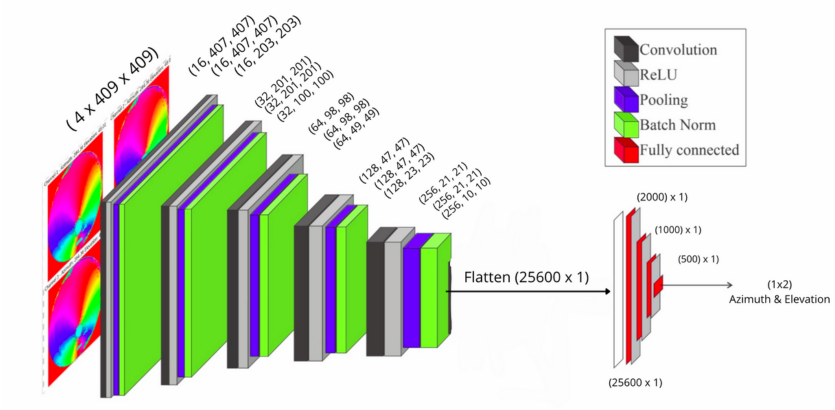
Robot Experiment to get Dataset (click to play the video)



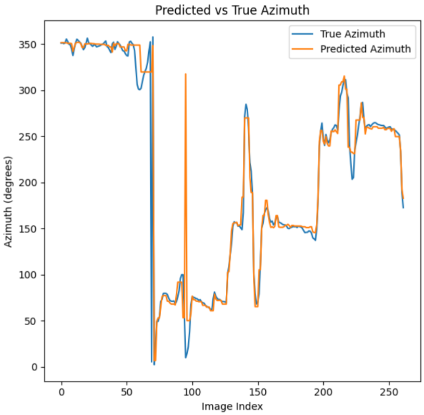
Sun Ephemeris



CNN Model for Machine Learning



Predicted vs Truth Azimuth



How we can use skylight polarization?

Goal: Go to direction North.

1. **Measure Sun Azimuth from AoP**

$$\text{Robot measurement: } \alpha_{\text{meas}} = 110^\circ$$

2. **Get True Sun Azimuth from Ephemeris**

$$\text{Ephemeris: } \alpha_{\text{true}} = 135^\circ$$

3. **Compute Heading Error**

$$\Delta\theta = \alpha_{\text{true}} - \alpha_{\text{meas}} = 135^\circ - 110^\circ = 25^\circ$$

4. **Interpretation**

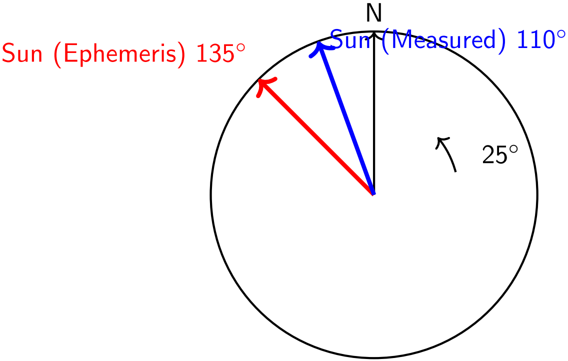
$$\Delta\theta = 25^\circ \quad \Rightarrow \quad \text{Robot is rotated 25° clockwise from North}$$

5. **Correction**

Rotate robot 25° to the left (anticlockwise)

$\Rightarrow$ Robot now faces True North

Sun Azimuth and Heading Error – Visual Explanation



- Red arrow = where the Sun **should** be (from ephemeris)
- Blue arrow = where the Sun **appears** from AoP measurement
- Angle between them = **heading error**
- Rotate robot **25° left** to face True North

Go Forward using AoP Corrections (every 30 min)

Goal: Travel forward while keeping an accurate absolute heading (True North) using periodic AoP-based sun azimuth corrections.

1. **Start / Initialize:**
   - Get accurate clock, initial estimated position (lat,lon) or start dead-reckon = known start.
2. **Move forward:** drive using IMU/odometry and maintain current heading.
3. **Every 30 minutes:**
   1. Capture AoP image.
   2. Estimate measured sun azimuth α_{meas} from AoP pattern.
   3. Compute predicted sun azimuth α_{true} = Ephemeris(lat_est, lon_est, time).
   4. Compute heading error (circular difference)

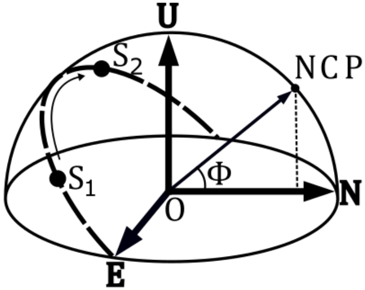
$$\Delta\theta = \text{wrap}\left(\alpha_{\text{true}} - \alpha_{\text{meas}}\right)$$

   5. Apply correction: rotate robot by $\Delta\theta$ (or update heading state in filter).
   6. Update estimated position using odometry and corrected heading.
4. **Repeat:** continue moving; next correction after 30 minutes.

Locating the Celestial Pole from Skylight Polarization

Core observation

During the day, the Sun appears to move in a circular path around the North Celestial Pole (NCP) due to Earth's rotation.

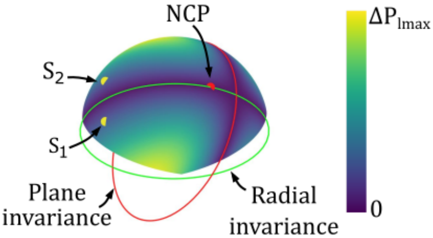


Symmetry of the DoLP pattern

Key discovery – Symmetry of the DoLP pattern

When comparing the Degree of Linear Polarization (DoLP) at two different times of the same day:

- DoLP changes over most of the sky
- **Two great circles remain perfectly unchanged** → called invariance circles (radial and plane symmetries)



The North Celestial Pole is special

The Most Important Thing

- The NCP lies on one of these invariance circles **every day**
- It is the only point in the entire sky with **nearly constant DoLP throughout the entire day**

SkyPole Principle – Finding the NCP from DoLP Variations

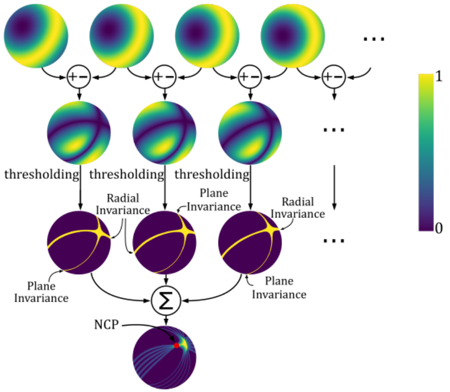
Simple idea:

Track which points do NOT change their DoLP over time

Practical method (3–4 images are sufficient):

1. Take several full-sky DoLP images during the day (hours apart)
2. Compute absolute differences between pairs $\rightarrow$ invariance circles appear as black lines
3. Sum all binary difference images $\rightarrow$ pixel voting
4. The pixel that remains invariant in all comparisons
   $\Rightarrow$ **intersection of all invariance circles**
   $\Rightarrow$ North Celestial Pole found!

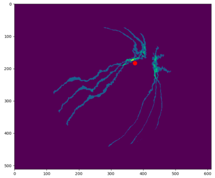
SkyPole Principle – Finding the NCP from DoLP Variations



Kronland-Martinet, et al.(2023). SkyPole—A method for locating the north celestial pole from skylight polarization patterns. Proceedings of the National Academy of Sciences, 120(30), e2304847120.

From NCP Position to Geographic Latitude

"As the NCP is located in the sky at an azimuth equal to that of the geographical north (i.e. **azimuth = 0°N**) and an altitude equal to the observer's latitude..."



So once SkyPole has found the red-dot pixel → we immediately know the pixel points toward the invisible North Celestial Pole and transform it from camera frame to ENU frame on latitude

Let's Practice It!

https://github.com/fauzikbr/polarization-course/